A Comparative Study of Extractive and Abstractive Text Summarization Techniques

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***Abstract*—Automatic text summarization reduces the effort required to consume long documents by producing shorter versions that preserve important information. This study presents a comparative experimental analysis of extractive and abstractive summarization techniques using the BBC News summary dataset, containing 2,225 news articles from five categories: business, entertainment, politics, sport, and technology. The study evaluates TF-IDF centroid summarization, TextRank, K-Means clustering, Agglomerative Hierarchical clustering, DBSCAN, and supervised sentence-selection approaches based on Gaussian Naïve Bayes, Decision Tree, and K-Nearest Neighbours. An abstractive BART model is additionally evaluated to compare neural generation with sentence-extraction approaches. The experimental pipeline includes text cleaning, sentence tokenization, stopword removal, lemmatization, TF-IDF representation, cosine-similarity features, sentence-level feature engineering, weak supervision, model training, and ROUGE-based evaluation. Among the evaluated extractive methods, TextRank obtained the strongest average performance, with ROUGE-1, ROUGE-2, and ROUGE-L F1 scores of 0.4128, 0.3955, 0.3517, respectively. BART achieved lower ROUGE scores on this dataset (0.2950, 0.1961, and 0.2249), while requiring substantially greater computational time. The results indicate that graph and similarity-based extractive methods can remain highly competitive when reference summaries are relatively extractive, whereas abstractive methods provide a different generation paradigm whose quality is not fully captured by lexical-overlap metrics alone.**

***Index Terms*—text summarization, extractive summarization, abstractive summarization, TF-IDF, TextRank, clustering, BART, ROUGE, natural language processing, data mining.**

I. Introduction

With the rapid growth of digital news, reports, and online documents, users increasingly face information overload. Reading every document in full is often impractical, motivating automatic text summarization systems that generate concise representations while preserving the most important information.

Text summarization can broadly be divided into extractive and abstractive approaches. Extractive systems select sentences or phrases directly from the source document, where abstractive systems generate new text that may paraphrase the source. Extractive methods are generally simpler, more interpretable, and computationally efficient, while abstractive methods can produce more fluent and compact summaries but require substantially more computational resources and careful evaluation.

This work performs a comparative data mining oriented study of several summarization strategies on the BBC News Summary dataset. The investigation covers classical sentence-ranking methods, graph-based ranking, clustering-based approaches, supervised sentence-selection models, and a transformer-based abstractive model. Rather than evaluating a single algorithm, the study examines how different modeling assumptions affect summary quality and computational cost.

The main objectives are: (i) To implement and compare multiple extractive and abstractive text summarization techniques using the BBC News Summary dataset and e valuate their effectiveness using ROUGE metrics; (ii) To analyze the trade-off between summarization quality and computational efficiency by comparing the performance, summary characteristics, and execution time of the evaluated techniques.

II. Related Work

Early automatic summarization research primarily relied on sentence scoring, term frequency, positional information, and other surface level linguistic features. Luhn proposed one of the earliest frequency-based approaches, while Edmundson introduced a broader framework for sentence selection using multiple indicative features [1], [2].

Graph-based summarization became influential through TextRank, which represents sentences as nodes and their semantic similarity as central to the document [3]. Such methods are attractive because they do not require task-specific supervised training.

Machine-learning approaches can formulate summarization as sentence classification or ranking. Features such as sentence length, position, TF-IDF similarity, lexical overlap, and structural properties can be used to predict whether a sentence should appear in a summary. Clustering methods provide another perspective by organizing sentences into group and selecting representative sentences from different regions of the document.

More recently, transformer architectures have enabled abstractive summarization. BART uses a denoising sequence-to-sequence pretraining onjective and has demonstrated strong performance on text generation and summarization tasks [4]. However, neural generation is computationally more expensive than conventional sentence-ranking methods.

ROUGE is one of the most widely used automatic evaluation families for summarization. ROUGE-1 and ROUGE-2 measure unigram and bigram overlap, while ROUGE-L is based on the longest common subsequence [5]. Although useful, lexical-overlap evaluation may favor summaries that use wording similar to the reference and not completely reflect semantic quality.

III. Methodology

A. System Overview

The proposed system provides a comparative framework for evaluating extractive and abstractive text summarization techniques. The BBC News Summary dataset containing 2,225 articles is first processed to prepare clean sentence-level text. TF-IDF and cosine similarity are then used to represent the sentence and extract relevant features. The processed data is passed to multiple summarization approaches. The generated summaries are evaluated against the reference summaries using ROUGE metrices, followed by performance and computational cost analysis.

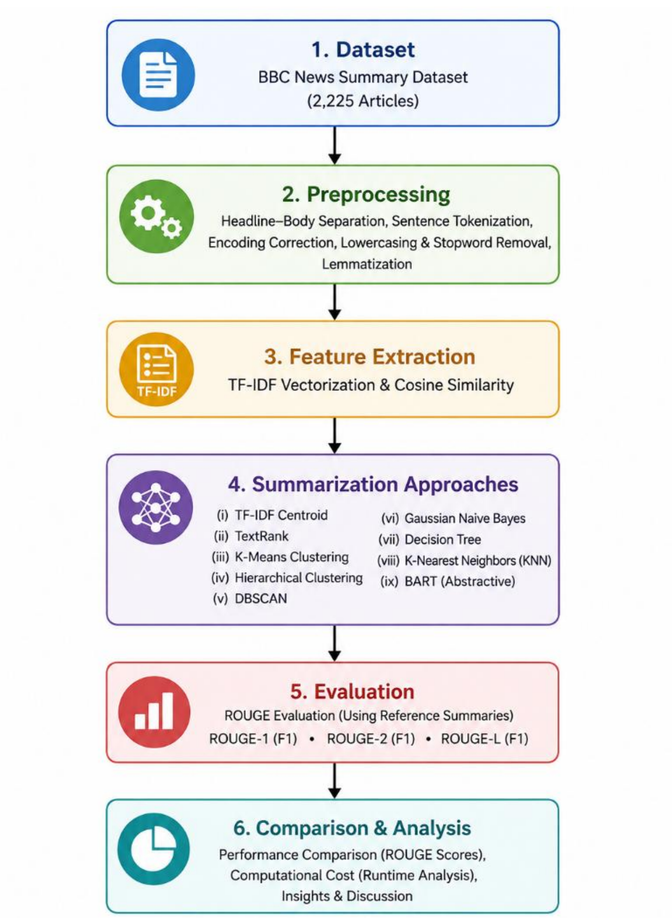

Fig. 1: System block diagram of the two-stage pipeline.

B. Dataset and Preprocessing

The study uses the BBC News Summary dataset, consisting of 2,225 news articles from five categories: business, entertainment, politics, sport, and technology. Each article contains its corresponding reference summary, which is used for evaluating the generated summaries.

The raw articles are converted into a suitable format for sentence-level summarization. The preprocessing stage includes headline-body separation, sentence tokenization, encoding, lowercasing, stopword removal, and lemmatization. Sentence tokenization divides each article into individual sentences, allowing the subsequent methods to rank, cluster, or classify sentences independently.

C. Feature Extraction

After preprocessing, TF-IDF vectorization is applied to represent the textual content numerically. TC-IDF assigns higher importance to term that are frequent in a particular document but less frequent across the complete corpus. Cosine similarity is then used to measure the similarity between sentence vectors. These representations are used by the centroid, graph-based, clustering, and supervised approaches.

D. Summarization Approaches

The proposed framework evaluates several summarization techniques:

1. TF-IDF Centroid: Sentences are scored according to their cosine similarity to the document centroid, and the most representative sentences are selected.
2. TextRank: A similarity graph is constructed in which sentences are represented as nodes. Sentences with higher graph-based importance are selected for the summary.
3. K-Means Clustering: Sentence TF-IDF vectors are divided into clusters representing different content groups, and representative sentences are selected from the clusters.
4. Agglomerative Hierarchical Clustering: Sentences initially form individual clusters, which are progressively merged according to their similarity.
5. DBSCAN: Sentences are grouped according to density rather than requiring a predefined number of clusters. Sparse sentences can be treated as noise.
6. Supervised Sentence Selection: Gaussian Naïve Bayes, Decision Tree, and K-Nearest Neighbors (KNN) are used to classify sentences for summary selection using the engineered sentence-level features.
7. BART: The pretrained bart-large-cnn model is used as the abstractive summarization approach. Unlike extractive methods, BART generates a new summary rather than directly selecting sentences from the articles.

E. Evaluation

The generated summaries are compared with the corresponding reference summaries using ROUGE-1, ROUGE-2, and ROUGE-L F1 scores. ROUGE-1 evaluates unigram overlap, ROUGE-2 evaluates bigram overlap, and ROUGE-L measures sequence-level similarity.

In addition to ROUGE evaluation, the study performs diagnostic analysis of reference-summary extractiveness, summary length, and BART performance with respect to reference-summary length. The project found that the reference summaries have varying degree of direct overlap with article sentences, which is important when interpreting the performance of extractive and abstractive approaches.

F. Performance and Computational Analysis

Finally, the summarization methods are compared based on their ROUGE performance and computational cost. Runtime analysis is used to examine the practical efficiency of classical extractive methods against the computationally more demanding abstractive BART model. The combined analysis provides a basis for selecting an appropriate summarization technique according to both summary quality and computational requirements.

IV. Results and Analysis

The methods were evaluated using ROUGE-1, ROUGE-2, and ROUGE-L F1 scores. These metrics measure lexical overlap between generated and reference summaries. The reported values are averaged over the evaluated article set.

TABLE I: ROUGE F1 Performance of the Main Summarization Methods.

Method	ROUGE-1 F1	ROUGE-2 F1	ROUGE-L F1
TD-IDF	0.409937	0.392536	0.349807
TextRank	0.412774	0.395473	0.351720
K-Means	0.360734	0.321341	0.312067
Hierarchical	0.366607	0.330510	0.317148
DBSCAN	0.403804	0.384398	0.344374
BART	0.295033	0.196127	0.224899

TextRank achieved the highest overall ROUGE scores among the evaluated methods. Its advantage over TF-IDF was modest but consistent across ROUGE-1, ROUGE-2 and ROUGE-L. DBSCAN also performed competitively, while K-Means and hierarchical clustering produced lower scores. The clustering methods may trade lexical similarity for broader sentence coverage, which is not always rewarded by ROUGE when the reference summary uses particular wording.

The BART model obtained lower ROUGE scores than the strongest extractive methods in this experiment. This result should be interpreted as proof that abstractive summarization is intrinsically inferior. The BBC reference summaries and the evaluation matric similar meaning with different wording. In addition, the experiment uses a pretrained BART model without task-specific fine-tuning on the complete dataset.

V. Computational Cost

Runtime was measured to compare the practical cost of the approaches. The following average processing time was obtained per article:

TABLE II: Reported Average Runtime Per Article.

Method	Average Time / Article
TD-IDF	0.43 ms
TextRank	2.44 ms
K-Means	74.53 ms
Hierarchical	1.31 ms
DBSCAN	8.69 ms
BART	611.11 ms

The runtime results reveal a major efficiency gap between conventional extractive methods and neural generation. TF-IDF and TextRank require only a few milliseconds per article in the reported setup, whereas BART requires hundreds of milliseconds per article. K-Means is also considerably more expensive than the simpler ranking approaches because of iterative clustering. Therefore, method selection depends not only on ROUGE quality but also on application latency and computational resources.

VI. Diagnostic Analysis

The project includes diagnostic analysis beyond the primary ROUGE table. Reference summaries were examined for their degree of extractiveness, and generated summaries were compared in terms of length and lexical overlap. These analysis help explain why extractive methods perform strongly on the selected benchmark.

The findings suggest that the reference summaries contain substantial overlap with source sentences. Consequently, methods that identify representative source sentences have a structural advantage under ROUGE evaluation. The result also highlights an important evaluation issue: a lower ROUGE score for an abstractive system down not necessarily imply lower human-perceived informativeness or fluency.

The classifier-based experiments further demonstrate the feasibility of treating summarization as a sentence-selection problem. However, weak supervision introduces noise because automatically assigned labels are proxies for the desired extractive decisions rather than expert annotations. This limitation should be considered when interpreting classifier performance.

VI. Conclusion and Future Work

This paper presented a comparative study of extractive and abstractive text summarization methods using the BBC News Summary dataset. The study established a common preprocessing and feature-engineering pipeline and evaluated centroid-based, graph-based, clustering-based, supervised, and transformer-based approaches. TextRank achieved the strongest reported ROUGE performance among the evaluated methods, closely followed by TF-IDF and DBSCAN. BART produced lower ROUGE scores in the present configuration while requiring substantially higher computational time.

The results show that classical data-mining and NLP methods remain effective for news summarization, particularly when reference summaries have strong extractive characteristics. At the same time, the study demonstrates the importance of evaluating abstractive systems with metrics that account for semantic similarity and generation quality rather than relying exclusively on lexical overlap.

Future work should investigate domain-specific fine-tuning of transformer summarizers, larger and more diverse datasets, semantic evaluation metrics such as BART score, human evaluation for coherence and factuality, and hybrid extractive-abstractive architectures. Additional experiments could also optimize clustering-based representative selection and compare stronger supervised ranking models.

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